
CIS437 - Cloud Computing

– Syllabus –

Instructor: Yong Zhuang

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Overview

- Welcome!
- Syllabus
- Cloud things

But first, let's play a "read and repeat" game!

**I WILL NOT
USE MY CREDIT CARD
ON ANY CLOUD SERVICE**

But first, let's play a "read and repeat" game!

EVEN IF THEY ASK NICELY

SO I CAN START A FREE TRIAL

What we will cover

Wide variety of cloud technologies

- Mostly focused on Google Cloud, but we'll take a look at AWS
 - Azure doesn't give out free education credits and I don't want you to burn your free trials
 - Fortunately, all skills are transferable between cloud environments
- All different points of the cloud stack

Knowledge you are assumed to have

Ability to program in **some language**

- You will be working on a real project!

Ability to work **individually** and on **teams**

- You're an adult, I also expect maturity and even effort-sharing where necessary

Ability to **talk to me if you have problems**

- Don't be nervous, I am very approachable!

How class will generally work

Our classes are 2 - 1 hr 15 min sessions per week

- Classes will be a mix of:
 - Lecture
 - Discussions
 - Labwork
 - Guided/self-guided work

Subject to change based on material, naturally

- But, I want you to get **practical experience**

Project



You will develop a half-semester-long project

Goals of this project:

1. Have something portfolio-worthy at the end
2. Develop it like a real-world project
 - a. Proposal, updates, presentation, etc.
3. More detail around midterm - start thinking of something *now* you'd like to "cloudify"

Project?

I want you to create something that is **interesting to you**

Aspiring app developer?

- Make that app (...using cloud technologies)!

Sysadmin?

- Manage a load-balanced server application of some sort!

Are you a software developer with latent tendencies for video game design?

- Make a *cloud-hosted* video game!

No clue?!?!

- Let's talk

It will need to contain a handful of cloud services though, not just one!

Our tech stack

Class is **synchronous in-person**, meaning:

- 1) Class runs at specific days/times
- 2) Office hours are in-person (can be online) as well

Class website: Blackboard // <https://gvsu-cis437.github.io/syllabus.html>

Async Chat: Discord

If you want to get a hold of me for questions:

- Ping me in Discord
- Email me
- Visit office hours (virtual or in-person (generally))

About this Course

Admin, Logistics, Grading

Contact Information

- ▶ Instructor: Dr. Yong Zhuang
- ▶ You can call me Dr. Zhuang (draw on), or Yong (you own)
- ▶ Homepage: <https://yong-zhuang.github.io>
- ▶ E-mail: yong.zhuang@gvsu.edu
- ▶ Office: MAK D-2-234
- ▶ Office hour: T Th 4:00 pm- 5:00 pm, MAK D-2-234
 - ▶ Or by appointment. Send me an email to schedule a time to meet (over Zoom or in-person).
 - ▶ <https://gvsu-edu.zoom.us/j/3966686420?pwd=WGxpc0N4YWcvOU9aWGxWZGYxbXZUdz09>



Textbook

Instructor's Lecture Notes, Open Educational Resources, and Handouts

Grading

- ▶ Quizzes & In-class Activities 15%
- ▶ Homework assignments 30%
- ▶ Project 25%
- ▶ Midterm 15%
- ▶ Final Exam 15%

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Grading: Homework

Homework: 30%

- ▶ Homework assignments are designed to reinforce course material and include a mix of programming tasks along with mathematical or written questions.
- ▶ Deadline: 11:59 pm Michigan time on the due date.
 - ▶ *Late policy*: Assignments submitted late will incur a 10% penalty per day, capped at 3 days (30%). After this period, the assignment **will not be accepted**.
- ▶ **No copying or sharing of homework!**
 - ▶ But you can discuss general challenges and ideas with others

Grading: Term Project

Course project: 25%

- ▶ Group project
- ▶ The term project is a half-semester long effort centered around cloud computing.
Grades will reflect the quality of the software created.
- ▶ Groups of 1–3 people will be accepted,

Academic Honesty

- ▶ Document all collaborations.
- ▶ No electronic code transfers between students.
- ▶ Code you find on the internet must be cited, with an active link to that code. That code should not solve the entirety of an assigned problem/project (i.e., don't have someone else do your project for you).
- ▶ You are encouraged to engage in conversations in online forums, but do not post solutions or solicit others to complete your work for you.
- ▶ You are encouraged to talk about problems with each other in non-technical terms (i.e., not code)
- ▶ Generative AI Tools: Unless an assignment states otherwise, generative AI tools may be used for explanation, brainstorming, and debugging. Any substantive use must be disclosed and cited. Students may not submit AI-generated code, analysis, or writing that they do not understand and cannot explain. Generative AI tools may not be used during quizzes or exams unless explicitly permitted. Students remain responsible for the correctness, originality, security, and academic integrity of all submitted work.
- ▶ Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.
 - ▶ **Suspicious cases will be reported to the Academic Honesty Committee**

Tentative Course Content

✂ September 6 - 7, Labor Day Recess: No classes!

✂ October 26-27, Fall Break: No classes!

✂ November 25-29, Thanksgiving Recess: No classes!

Module	Topics Covered
1	Syllabus, Introduction
2	Cloud Platforms
3	Cloud Shell, Cloud Applications
4	* as a Service, Microservices, Serverless Computing
5	Security and Geographic Implications
6	Various Sizes of Data
7	APIs
8	Cloud Run, Cloud Build
9	Cloud Operations
10	Machine Learning and Generative AI
11	Midterm Exam (October 15, 2:30 PM - 3:45 PM)
12	Final Exam (December 17, 2:00 PM–3:50 PM)

Expectations for Success

- ▶ Check Blackboard on a regular basis for announcements and assignments
- ▶ Check <https://gvsu-cis437.github.io> for course material.
- ▶ I'm here to help you! Come to office hours or schedule an appointment when you're feeling lost/stuck.
- ▶ Let me know when something is getting in the way of your success in this class.
- ▶ Let me know how the class and my teaching can be improved.
- ▶ Adhere to class, CIS, and GVSU policies on academic honesty.

Resources

GVSU provides opportunities for students to improve your academic skills through resources, such as

- ▶ Writing center (<https://www.gvsu.edu/wc/>)
- ▶ Speech lab (<https://www.gvsu.edu/speechlab/>)
- ▶ Research consultants (<https://www.gvsu.edu/csce/>)
- ▶ Library liaisons (<https://www.gvsu.edu/library/about-the-university-libraries-3.html>)
- ▶ Academic success center (<https://www.gvsu.edu/sasc>)

Warming Up

Individual introduction

- Name and what do you want to be called
- Describe your background and experience in Cloud Computing.
- Specific topics you seek to learn from this class.

About Me

- You can call me Dr. Zhuang (draw on), or Yong (you own)
- Education:
 - Ph.D., M.S. in Computer Science from the University of Massachusetts
- Experience/Interests:
 - My research interests include web application design, machine learning, and data mining.
- Personal Interests:
 - I enjoy spending time in city parks, hiking, and playing board games with friends and family.

